

RONAK CHATURVEDI

AI Developer | Generative AI, Local LLM Systems & RAG

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SUMMARY

AI Developer specializing in Generative AI, local LLM production systems, and Retrieval-Augmented Generation. Ships production LLM automation at GreenPoint Global — a LoRA fine-tuned Ollama pipeline generating board-report commentary from PDFs (live daily), an LLM-to-code Manim video system, and a self-hosted RAG platform with verified page-level citations. Strong in local model deployment, prompt architecture, grounded structured output, and anti-hallucination guardrails.

TECHNICAL SKILLS

Languages: Python, SQL (T-SQL), TypeScript

Generative AI & LLM: Ollama (local LLM deployment), Hugging Face Transformers, LoRA fine-tuning, Prompt Engineering, Prompt Architecture, JSON-schema-constrained generation, Local multimodal / vision inference, Anti-hallucination guardrails

RAG & Retrieval: Hybrid retrieval (full-text + semantic), pgvector, sentence-transformers, Embeddings, Cross-encoder reranking, Chunking, Grounded citation verification

Backend & Frontend: FastAPI, REST APIs, Next.js / React, Streamlit

ML & Data Processing: PyTorch, torchvision, scikit-learn, NumPy, Pandas, pdfplumber, PyMuPDF (fitz), regex-based parsing, Grad-CAM

Automation & Low-Code: Power Automate (cloud & desktop flows), Power Apps, AI Builder

Deployment & Tooling: Docker, Docker Compose (multi-stage builds), Git / GitHub, Manim (Community Edition), AST-based static code validation

Databases: PostgreSQL (+ pgvector), Microsoft SQL Server (mssql-python, T-SQL, stored procedures)

PROFESSIONAL EXPERIENCE

AI Developer, GreenPoint Global Jun 2026 – Present

AI Financial Commentary Engine (Credit Unions) | Python, Ollama, LoRA, pdfplumber, SQL Server

- Built and deployed an end-to-end Python pipeline converting credit union board report PDFs (Power BI exports) into executive-level, LLM-generated CEO commentary across 9 sections plus synthesized CEO Summary and Recommendations — live in daily production, processing ~50 PDFs in ~7 hours with no manual drafting.
- Replaced unreliable image-only LLM extraction with a deterministic pdfplumber/PyMuPDF extraction layer (regex table/row parsing, denomination-scale detection, section alias mapping) producing code-verified figures for most sections.
- Designed a multi-layer validation and guardrails system catching segment-sum mismatches, mislabeled data, and cross-source conflicts (Shares vs. Balance Sheet), using a fail-closed drop-and-flag pattern before data reaches the LLM.
- Engineered a modular prompt architecture (9+ section prompts over shared rules for voice, numeric integrity, and evidence discipline), iterating through five versions to eliminate JSON-format conflicts, numeric hallucination, and unsupported interpretive language.
- Added anti-hallucination safeguards (n-gram overlap detection, JSON salvage/repair), a hybrid code + vision-model extraction strategy with reconciliation, a LoRA fine-tuned Ollama model, and a SQL Server persistence layer (T-SQL, stored procedures) for Power BI.

Automated Math Education Video Pipeline | Python, Manim, Ollama, AST validation

- Shipped a local LLM-to-code system converting plain-text lesson scripts into rendered educational videos — 3-minute grade 9–10 videos in under 10 minutes each, 10+ shipped to production, owned solo end-to-end.
- Designed content-based routing: LLM-generated Manim scene functions for narration, deterministic hand-written templates for any scene with real formulas — guaranteeing math renders verbatim, never paraphrased or shrunk to fit.
- Built a two-layer safety net (AST-based static validation before caching plus runtime template fallback) with content-hashed scene caching, so bad LLM output degrades gracefully and re-renders skip redundant inference.
- Diagnosed a content-fidelity failure where the local LLM silently paraphrased formulas, and a parsing gap dropping formula blocks — tracing each from a rendered artifact back to the rule at fault.

Local RAG Platform for PDF Document Intelligence | FastAPI, Next.js, PostgreSQL/pgvector, Ollama, Docker

- Built a full-stack, self-hosted Retrieval-Augmented Generation application (FastAPI, Next.js/React, PostgreSQL + pgvector, containerized Ollama) for querying PDFs with grounded, page-level citations — delivered as a client handoff, not a prototype.
- Implemented hybrid retrieval combining keyword (full-text) and semantic (embedding) search with a cross-encoder reranking stage filtering topically-similar-but-irrelevant results before generation.
- Enforced strict grounding so the model cannot invent citations — every source is verified against retrieved chunks — with a page-level viewer rendering the exact PDF page on demand (PyMuPDF), plus database-layer category filtering.
- Placed every external dependency (storage, embeddings, LLM) behind swappable provider interfaces so components swap via config with no code changes; containerized the 5-service stack via Docker Compose with multi-stage builds, shipping pre-built images and setup docs.

Reporting Automation — Credit Union SIO Pipeline | Power Automate, Power Apps, AI Builder, SQL Server

- Engineered production reporting automation with Power Automate (cloud and desktop flows), Power Apps, and AI Builder — including a Python SQL Server client (mssql-python) with certificate handling and MERGE-based upserts, plus DOCX-level OOXML fixes for table pagination and logo injection.

PROJECTS

CXR-Detect — Chest X-Ray Pneumonia Classifier | PyTorch, torchvision, scikit-learn, Streamlit2024

- Fine-tuned ResNet50 on 5,800+ chest X-rays using NumPy/Pandas data preparation; corrected a 3:1 class imbalance via WeightedRandomSampler, achieving 98.5% AUC, 96% accuracy, 98.2% recall, and 96.9% F1 on a 624-image test set.
- Halved training time with PyTorch AMP, cosine annealing LR, and early stopping; applied 5-pass test-time augmentation and Brier-score calibration, cutting misclassifications to 24/624; added Grad-CAM heatmaps and deployed as a Streamlit app with a tuned clinical decision threshold.

EDUCATION

Bachelor of Engineering (B.E.), Computer Science — AI/ML, with Honours in Data Science

Vishwaniketan Institute of Management Entrepreneurship and Engineering Technology (VIMEET), Maharashtra | CGPA: 8.4 | 2022 – 2026

ACHIEVEMENTS & PUBLICATIONS

- Ranked 1st in Honours (Data Science) with a CGPA of 9.2; Top-3 in core academics and Honours in 2nd and 3rd year.
- Publication: "Application for Direct Market Access for Farmer" — accepted, JETIR (International Journal).